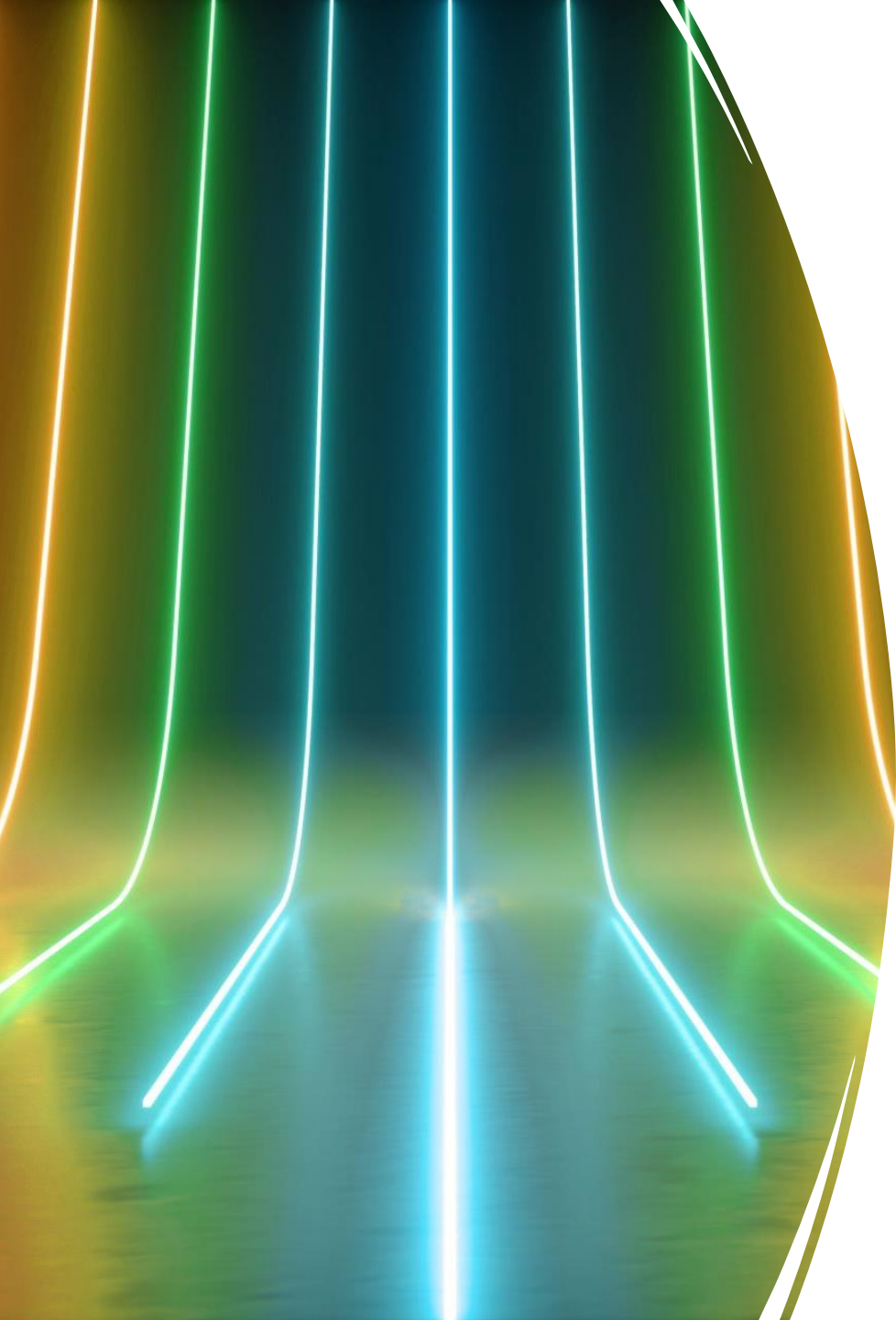




Healthcare AI Companion: Symptom checker, drug advisor & health insights

Hakan Guerler



PROBLEM

- Patients often struggle with understanding symptoms, managing medications, and staying updated on diseases.

DEVELOPING AN INTERACTIVE TOOL



DIEASE TO SYMPTOM PREDICTOR



DRUG INTERACTION ADVISOR



DISEASE NEWS FINDER

SYMPTOM-BASED DISEASE PREDICTOR

Healthcare Dataset (220 disease, 450k symptoms)

Dataset: Mendeley Data, Hugging Face, GTS AI

Data cleansing, merging,

Feature Engineering, Label encoding, Training (Accuracy: 0.95)

Streamlit

DRUG INTERACTION ADVISER



The patient prefers not to disclose the medications she is currently taking but wants the ability to independently check for potential interactions.



Meanwhile, the doctor already has access to the patient's medical history and aims to verify whether the newly prescribed medication might interact with the patient's regular medications.

DRUG INTERACTION ADVISER

- Data set: 47k Drugs
- Dataset: Drugbank, Mendeley Data
- Data: Cleaning, Merging, Classifying, Filtering
- Streamlit



DISEASE NEWS FINDER



Step 1: Fetching Related Articles

Sends a GET request to the NewsAPI to search for articles based on a user-provided query

Scrapes the full content of an article from its URL using BeautifulSoup



Step 2: Summarize Articles

Uses the Hugging Face summarization pipeline to summarize the content of fetched articles.

Initializes the summarizer pipeline with the pre-trained facebook/bart-large-cnn model.



Streamlit App

USE-CASE

- Patient: vomiting, indigestion, loss_of_appetite, passage_of_gases, internal_itching
- Doctors Diagnosis: Migraine
- Doctors prescription: Sulpiride / Digoxin
- News Finder: Recent research about Migraine

FUTURE PERSPECTIVE



Disease prediction

Scraping information on rare diseases from publications
Incorporating additional diagnostic data, including MRI, CT scans, and blood test results



Drug interaction

Incorporating a list of foods and minerals that may interact with drugs.



News Finder

Enhancing the system to find and summarize recent scientific publications about the disease, providing both doctor-focused insights and patient-friendly explanations.

ACKNOWLEDGMENTS



Thank you for listening



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